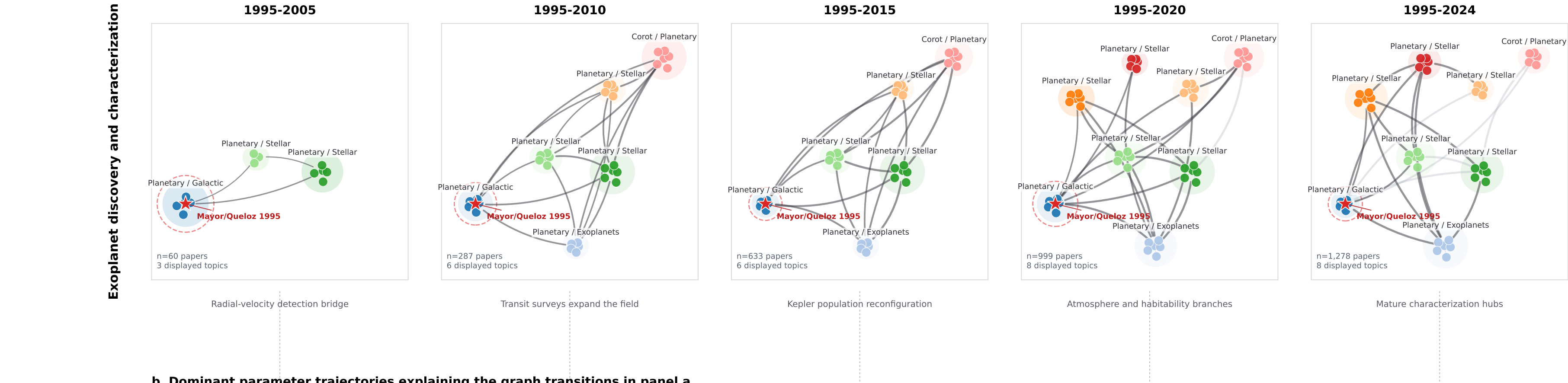


Fig. 1 | Citation knowledge-graphperturbation after a landmark innovation

Example domain: exoplanet discovery and characterization. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



b Dominant parameter trajectories explaining the graph transitions in panel a

